

Steve Butterworth

Product Engineer · Tech Lead

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PROFILE

Product engineer and tech lead specialising in data-heavy products, with 20+ years taking raw data from ingestion to the screen a customer uses. Built OLAP and analytics SDKs at Crystal Decisions and i2, live sports data systems for NTT DATA fed by GPS, timing and scoring feeds over Kafka and WebSockets, search alerts that have sent millions of emails, and LLM classification pipelines in production. More than 20 applications shipped to production for long-standing clients along the way.

Tech lead: led small engineering teams to immovable live-event deadlines, setting coding standards and reviewing every pull request. **Product-focused:** founded two products, and own work from the first customer conversation to long-term production. **Test-first:** coding agents now write most of my code, under guardrails I built for test-first development, code review, and security and performance audits. That leaves my time for architecture, reliability and tech debt: I kept one platform healthy for 13 years, from Rails 1 to Rails 8.

SKILLS

- **Engineering:** Architecture and data modelling, test-first development, CI, code review, security scanning, performance and reliability work, tech-debt management, small-team leadership, stakeholder communication.
- **Data pipelines:** Ingestion over Kafka, WebSockets, XML and third-party APIs; browser automation and scraping; enrichment; LLM classification and summarisation.
- **Search and storage:** PostgreSQL, Elasticsearch, Redis.
- **Visualisation:** D3.js, Three.js, dashboards, reporting, PDF and PowerPoint exports.
- **Application:** Ruby on Rails, Stimulus, Turbo, Tailwind, API design, Sidekiq, RSpec.
- **AI:** Classification and summarisation pipelines and tool-calling AI agents in production Rails apps; agentic development with Claude Code and Codex.

EXPERIENCE

Product Engineer, Environmentjob.co.uk (Contract)

2013 - Present

- Long-running retained contract on the UK's leading environmental jobs platform, held alongside other client work since 2013.
- Kept the platform current for 13 years, from Rails 1 to Rails 8, paying down tech debt alongside feature work.
- Build and maintain systems for job search, advertiser and job-seeker tools, admin workflow, CRM and reporting. Changes ship in days and stay in production for years.
- Designed the data model and built Elasticsearch full-text search, faceted filtering and an email alert system that has sent millions of emails.
- Built a software tool for monitoring relevant new job postings across hundreds of websites, later spun out as its own SaaS.

Founder & Product Engineer, Changeflow

2023 - 2026

- Founded, built and operate a bootstrapped Rails SaaS for website monitoring and change alerting, used by Am Law 200 firms.
- Designed the core data model, monitoring pipeline, background jobs and notification workflows end to end.
- Integrated Puppeteer/Node.js browser automation with Rails, PostgreSQL, Redis and Faktory/Sidekiq.
- Built LLM pipelines for classification, summarisation and cleaning, plus an in-app AI agent.
- Built a daily automated audit that flags suspected missed alerts, because regulated-industry customers cannot afford a silent miss.
- Now a stable, lightly maintained side product rather than an active venture.

Lead Engineer, Live Data Systems, LEX (Contract)

2019 - 2023

- Delivered time-critical data-ingestion and visualisation systems for NTT DATA sponsorships including The Open, Tour de France, BMW International Open and Indy 500, ingesting GPS, timing, weather and scoring feeds over Kafka, WebSockets and XML.
- Built ShotView, live 3D shot tracking in Three.js for The Open's website, taken from creative concept to over 1 million visits a day within 4 months; won Best Use of Technology at the Sports Industry Awards 2023.
- Rebuilt the NTT DataWall as a 30-metre 3D visualisation synced with the live broadcast, using real-time OCR of the feed and ActionCable to put matching player and hole analysis on screen within milliseconds.
- Led a small team of software engineers within a wider multidisciplinary team of creative directors, designers and producers, owning coding standards, automated testing and code review.

Product Engineer, Data Analytics, Retechnica (Contract)

2019

- Built a medical voice-of-customer analytics platform for GlaxoSmithKline (GSK), with data exploration, interactive visualisation and PDF/PPT reporting exports.
- Shipped on time and on budget with a 9/10 satisfaction score from 50+ stakeholders.

Senior Engineer, Live Data Experiences, Pulse Group (Contract)

2013 - 2018

- Developed the first generation of NTT DATA's live onsite data experiences and event websites in Rails, JavaScript and D3.js for The Open Championship.
- Delivered large-format live data screens for Team GB at the Olympic Games.
- Built Rails reporting tools, touch-screen interactives and a Cordova mobile events app for Emirates Airline.
- Recruited and managed other engineers as projects required, reviewing every pull request and running estimation and delivery against fixed event dates.

Founder & Product Engineer, Flumes

2010 - 2013

- Built a social media analytics platform from scratch, covering live social data ingestion, enrichment and interactive visualisation.

Full-Stack Rails Engineer, AlphaSights

2008 - 2010

- Joined as first employee, set the technology direction with the founders, and recruited and led the initial Ruby on Rails engineering team.
- Architected and built a bespoke CRM and customer-portal system underpinning the advisory firm's client communications.
- Set the company's software development practices from the start, including coding standards, unit testing, continuous integration and Kanban.

ADDITIONAL

- **Co-Founder, Unicorn Studios Ipswich** (2019 - Present): a not-for-profit creative co-working space for remote workers, freelancers and contractors.
- **Independent Software Engineer, Humble Technologies** (2005 - 2008): PHP and Ruby consultancy for UK clients.
- **Senior Software Engineer, i2, Inc. (now part of IBM)** (2002 - 2005): responsible for the web SDK of the company's flagship analytics tool; built on Java and JavaScript, it let third parties embed its charting technology in any web page.
- **Software Developer, Crystal Decisions (now part of SAP)** (1999 - 2001): built enterprise OLAP components and a front-end JavaScript SDK.

EDUCATION

BSc (Hons) Computer Science, 1st Class, University of Manchester